National University of Computer & Emerging Sciences, Karachi

Fall 2026 (CS)

Assignment # 1

Course Code: CS-1002	Course Name: Programming Fundamentals
Open Date: 7/September/2026	Deadline: 16/September/2026

Important instructions:

- You must never share your assignment with anyone. If any part of your assignment is found to be copied, then you will be awarded 0 marks as per the university policy.
- The assignment must be submitted before the deadline.
- Carefully read the submission instructions given below.

Submission:

- Each student should submit a single PDF containing all questions in order, named "A1-[StudentID].pdf".
- Each submitted file should have the STUDENT ID and NAME of the student at the top.
- All submissions will be made on Google Classroom.
- A random student may be called for VIVA on the submitted assignment; any cheating will be penalised by zero grades.
- The assignment will not receive any credit if submitted after the due date.
- Zero grade for plagiarism (copying/cheating) and for late submissions.

General Instructions

1. Attempt all questions in both Part A and Part B. Part A consists of foundational practice problems; Part B consists of complete system-design scenarios.
2. For every question in Part B, follow the design process specified within that question before writing any code do not skip directly to pseudocode or a flowchart.
3. All decision conditions, loop bounds, and nested structures must be written out explicitly in your algorithm, flowchart, and pseudocode. A step that hides the underlying logic inside a single vague sentence will not receive credit for that step.
4. If any part of a scenario appears ambiguous, state the assumption you have made clearly at the top of your answer for that question, and proceed on that basis.
5. Handwritten diagrams are acceptable if scanned clearly and inserted in the correct place in the document; illegible or missing diagrams will not be marked.

Part A — Foundational Practice Problems

The problems in this part are shorter and more direct than the system-design scenarios in Part B. Each one targets a specific decision-making or looping construct. Unless a question states otherwise, submit the algorithm and pseudocode.

Question 1

Take a number N and print numbers from N down to 1.

Question 2

Input a year and determine whether it is a leap year. (Rule: divisible by 4 AND (not divisible by 100 OR divisible by 400).)

Question 3

Take a number, count how many digits it has, and state whether it is a single-digit, double-digit, or triple-digit (or more) number.

Question 4

Print the multiplication tables from 1 to 10, but skip the table of any number that is a multiple of 3.

Question 5

Take a number, repeatedly sum its digits until the result is a single digit, and display each intermediate result. For example, if the number is 123, the sum of its digits is 6.

Part B System Design Scenarios

The scenarios in this section describe complete systems rather than isolated calculations. Each question presents its own specific requirements; therefore, read every scenario carefully, as the requirements may vary from one question to another.

For each question, students must submit the following:

- Algorithm
- Problem Analysis Chart (PAC)
- IPO Chart
- Flowchart
- Pseudocode
- C Program

The program implementation must focus on decision-making statements, such as if, if-else, nested if, and switch statements, wherever appropriate.

Some real-life scenarios may naturally require repetition or processing multiple records. However, since the purpose of this assignment is to practice decision-making statements, do not implement loops or repetition structures at this stage. Consider only a single input and output-based execution of the system.

The focus of this assignment is to analyze a complete scenario and implement its programming logic using appropriate decision-making statements.

Question 1 Hotel Booking System

A hotel processes booking requests for N guests. For each guest, take the season (Peak or Off-Peak) and room type (Standard, Deluxe, or Suite) as input, and apply nested pricing logic: Peak season charges a higher base rate per room type, and Off-Peak charges a lower rate. In addition, if a guest stays more than 7 nights, a long-stay discount applies regardless of season. Display the final price for each guest and the hotel's total revenue after all guests have been processed.

- The loop runs for N guests (for example, N = 4).
- Peak season rates: Standard = Rs. 5,000/night, Deluxe = Rs. 8,000/night, Suite = Rs. 12,000/night.
- Off-Peak rates: Standard = Rs. 3,000/night, Deluxe = Rs. 5,000/night, Suite = Rs. 8,000/night.
- If nights stayed exceed 7, apply a flat 15 percent long-stay discount to the total, regardless of season.
- Total price = (rate × nights) − discount, where applicable.
- Accumulate every guest's total into a running Hotel Total Revenue figure, displayed after the loop ends.

Question 02 Elevator Simulation

An elevator starts at the ground floor and receives a list of floor requests, processed one at a time in a loop. For each request, compare it to the elevator's current floor: if the requested floor is higher, print "Moving Up"; if lower, print "Moving Down"; if the same, print "Doors Opening." Update the current floor after each stop.

- The elevator starts at Floor 0.
- The loop processes a list of N floor requests (for example, N = 5 requests: 3, 1, 5, 2, 2).
- Requested floor greater than current floor → "Moving Up".
- Requested floor less than current floor → "Moving Down".
- Requested floor equal to current floor → "Doors Opening".
- Update the current floor to the requested floor after each stop.

Question 03 Class Result Processing

A school processes results for a class of N students. Each student has 5 subject marks, which must be summed and averaged. After computing the average, classify the student: average of 80 or above is "Distinction", 60 or above (but below 80) is "Pass", and below 60 is "Fail". In addition, if the student failed any single subject (a mark below 33), override the result to "Fail — Subject Deficiency", even if the average alone would otherwise pass.

- The loop runs for N students (for example, N = 3), each with 5 subject marks out of 100, processed by an inner loop.
- Average = sum of the 5 marks divided by 5.
- Average of 80 or above → "Distinction".
- Average of 60 or above and below 80 → "Pass".
- Average below 60 → "Fail".
- Override: if any single subject mark is below 33, the result becomes "Fail — Subject Deficiency", even if the average is 60 or above.

Question 04 Online Shopping Bill Calculator

An online shopping company wants to develop a simple Shopping Bill Calculator for its customers. When a customer purchases products, the system calculates the final amount in several stages. The calculations are dependent on one another: the result produced by one function is used by the next function.

Mathematical Functions Used

The system uses the following three mathematical functions.

- Subtotal: $s = q \times p$, where q is the quantity purchased and p is the price per item.
- Discounted Amount$=s - (s \times d) / 100$, where s is the subtotal.
- Final Bill$= a + (a \times t) / 100$, where a is the discounted amount and t is the tax percentage.

Requirement:

- Ask the customer to enter the quantity of products purchased.
- Ask for the price per item.
- Ask for the discount percentage.
- Ask for the tax percentage.
- Validate the entered values; if any value is invalid, display an appropriate error message and terminate the calculation.
- Store the calculation details, generate a bill document, and display the final bill to the customer before ending the program.

Question 05 Smart Campus Parking and Access Management System

A university is developing a Smart Campus Parking and Access Management System to control vehicle entry, assign parking spaces, and monitor parking capacity during the day. The university has three parking zones: Zone A (Faculty) with a maximum capacity of 20 vehicles, Zone B (Students) with a maximum capacity of 40 vehicles, and Zone C (Visitors) with a maximum capacity of 15 vehicles. The security officer enters the number of vehicles that are expected to arrive during the current operating period. The system must process each vehicle one at a time and ask for its vehicle type (C for car, B for bike, or V for van), user category (F for faculty, S for student, or G for visitor/guest), and whether the vehicle has a valid parking permit (Y or N). The system must validate every entered value before processing the vehicle. Invalid vehicle types, categories, or permit values must be rejected and the user must be asked to enter the information again. A vehicle with an invalid combination of information must not be counted as successfully processed.

For each valid vehicle, the system must determine its parking eligibility according to the following rules.

- Faculty members with a valid permit may park in Zone A.
- Students with a valid permit may park in Zone B.
- Visitors with a valid visitor permit may park in Zone C.
- A faculty member arriving in a van may use Zone A only if space is available.
- Student arriving in a van must be redirected to Zone C if the visitor zone has available space; otherwise, the vehicle is rejected.
- Bikes belonging to students with a valid permit may also use Zone B.
- Bikes belonging to faculty members with a valid permit may use Zone A.
- Visitors arriving by car or bike with a valid permit may use Zone C.
- Visitors arriving by van may use Zone C only when at least two spaces are available, because a van occupies two parking spaces.
- Any vehicle without a valid permit must be rejected unless it is an emergency vehicle; therefore, the system must additionally ask whether the vehicle is an emergency vehicle (Y or N). An emergency vehicle may enter regardless of its permit status, but it must be assigned to the appropriate zone based on the user category.

- A van assigned to Zone C consumes two parking spaces, whereas every car or bike consumes one space.

The system must continuously maintain the number of occupied spaces in each zone. Before assigning a vehicle, it must check whether sufficient capacity is available. If the required capacity is not available in the preferred zone, the system should apply the alternative-zone rules where applicable; otherwise, the vehicle must be rejected. For every successfully parked vehicle, the system should display the assigned zone and the remaining capacity of that zone. For every rejected vehicle, the system should display the reason for rejection, such as invalid permit, no available space, or no suitable zone. The system must also maintain counters for successfully parked cars, bikes, and vans, as well as rejected vehicles. Because multiple vehicles are processed, the program must use a loop, and because the system must repeatedly validate information and may need to examine the available zones before making an assignment, students are expected to use nested decisions, nested loops, and appropriate combinations of decisions and loops.

Requirement:

Show parking summary which includes:

- The total number of vehicles processed,
- Total accepted vehicles, total rejected vehicles,
- the number of cars, bikes,
- vans successfully parked, and the final occupancy and
- remaining capacity of each parking zone.
- Display the zone with the highest occupancy and indicate whether the entire campus parking facility is full.

Question 06 Smart EV Charging and Parking Management System

An urban shopping mall is developing a Smart EV Charging and Parking Management System. When an electric vehicle arrives, the system must determine whether to admit it, which charging category to assign, and how much the customer should pay.
The system asks the driver for:

- Vehicle type: E = Electric Vehicle, H = Hybrid Vehicle.
- Battery charge level (SOC) as a percentage.
- Required charging level as a percentage.
- Expected parking duration in hours.
- Current time in 24-hour format.
- Whether the customer has a parking membership (Y/N).
- Whether the customer has disabled-person priority status (Y/N).
- Whether the charging station is currently available (Y/N).

The system should make decisions according to the following rules.

- If the charging station is unavailable: if the vehicle is a hybrid, display "Charging unavailable – Parking only." Otherwise, display "No charging slot available." If the station is available, continue with the remaining checks.
- A fully electric vehicle (E) can use the charging station. A hybrid vehicle (H) can use the charging station only if its battery level is below 40 percent. Otherwise, display "Vehicle does not qualify for EV charging.
- Calculate Required Charging = Required Charging Level − Current Battery Level. If the required charging level is less than or equal to the current battery level, display "No charging required." Otherwise, continue.
- The system assigns a priority category, in the order given below.
    - Priority 1 — Emergency Charging: if battery level is 15 percent or below, and required charging level is 80 percent or above, assign "Emergency Charging Priority."
    - Priority 2 — Priority Customer: if the vehicle does not satisfy the emergency condition, but disabled-person priority is Y, or membership is Y and battery level is 30 percent or below, assign "Priority Charging."
    - Priority 3 — Normal Charging: otherwise, assign "Normal Charging."

- The mall uses different charging rates.
    - Off-peak (before 5 PM or after 10 PM): Rs. 35 per charging unit. If the customer is a member, they receive a 20 percent discount during off-peak hours.
    - Peak (5 PM to 10 PM): Rs. 50 per charging unit, with a 10 percent discount during peak hours.
    - No membership discount is provided for Emergency Charging Priority.
- Parking charges are calculated separately from charging charges.
    - First 2 hours: Rs. 200.
    - More than 2 and up to 5 hours: Rs. 400.
    - More than 5 hours: Rs. 700.
- A member receives an additional 20 percent parking discount. Disabled-person priority customers receive free parking. Free parking applies only to parking charges, not to electricity/charging charges.
- If the expected parking duration is more than 8 hours, display "Long-stay warning: Please relocate your vehicle after charging." Otherwise, display "Standard parking duration."

Requirements:

The system should display:

- Vehicle type
- current battery percentage
- Required charging percentage
- Charging priority, peak/off-peak status
- Charging cost, parking cost, discount,

- Final payable amount
- Appropriate warning/message.